

# IRTForests

Andrew T. Scott, Fall 2025

[github.com/ascott02/IRTForests](https://github.com/ascott02/IRTForests)

# Item Response Theory + Random Forests

- Trees become respondents, images become items.
- Response matrix records per-tree correctness on held-out examples.
- Goal: explain RF behavior via IRT ability & difficulty signals and vice versa.

# Motivation

- Fascinated by **IRT** for its **interpretative power**
- Meaning emerges only through a **cohort** — interactions matter
- Reminds me of **ensemble methods**, esp. **Random Forests**
- Both derive insight from **collective behavior**
- **Hypothesis:** Random Forests ↔ IRT — shared latent structure

# Agenda

- Background: IRT + RF primers
- Pipeline: datasets, embeddings, and response matrices
- Case studies: Rasch Family models (1PL) on CIFAR (PCA), CIFAR (MobileNet), MNIST
- Cross-study comparison, 2PL/3PL updates, takeaways, next steps

# Item Response Theory (IRT) (Wilson, 2005)

Why? Because performance  $\neq$  ability — but they're related.

- Classical Test Theory (CTT) tells us *how someone did on this test*.
- IRT models *how someone would perform on any set of items that measure the same underlying ability*.
- IRT doesn't replace CTT, it generalizes it with **portable, interpretable measurements** of capability.

| CTT                             | IRT                                          |
|---------------------------------|----------------------------------------------|
| Measures perf. on specific test | Estimates underlying ability                 |
| Test = sample of items          | Items = samples from a calibrated continuum  |
| Precision assumed constant      | Precision varies with ability                |
| Great for grading               | Great for understanding and interpretability |

A joint calibration framework where ability and difficulty are inferred together, each defined only in relation to the other.

It's less like grading individuals and more like synchronizing clocks — each calibrated against the ensemble.

# Item Response Theory Building Blocks

## Core Terms

- Ability ( $\theta$ ): respondent skill; higher  $\rightarrow$  higher success odds (1PL).
- Difficulty ( $\delta$ ): item hardness; higher  $\rightarrow$  harder even for strong respondents (1PL).
- Discrimination ( $a$ ): slope near  $\delta$  (2PL).
- Guessing ( $c$ ): floor for multiple-choice exams (3PL).

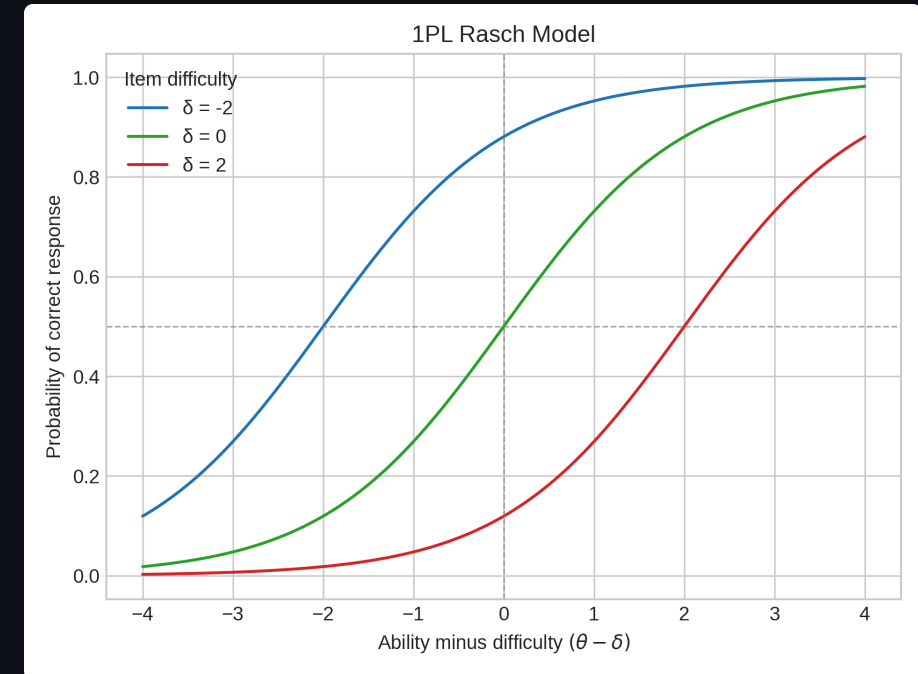
## Forest Analogy

- Respondents  $\rightarrow$  decision trees on a shared test set.
- Items  $\rightarrow$  images; responses are binary (tree correct?).
- Response matrix  $R_{ij} \in \{0, 1\}$  feeds variational IRT.
- Outputs: posteriors over  $\theta_i$ ,  $\delta_j$ , and information curves.

# Rasch (1PL) Model in One Picture

$$\Pr(R_{ij} = 1 \mid \theta_i, \delta_j) = \frac{1}{1 + e^{-(\theta_i - \delta_j)}}$$

- The probability a respondent gets the item correct, given their ability, and the item's difficulty.
- Single global slope keeps parameters on a shared logit scale.
- $\theta - \delta = 0 \Rightarrow 50\%$  success; shifts left/right change odds.
- Fisher information peaks where curves are steepest.
- See [IRT ICC Visualizer](#) for 2PL, 3PL, 4PL



1PL Item Characteristic Curves (ICC)

## IRT Output

- **Ability histograms** flag low-skill trees worth pruning.
- **Difficulty ladders** highlight mislabeled or ambiguous items.
- **Wright maps** overlay  $\theta$  and  $\delta$  to expose coverage gaps.
- **Information curves** reveal where ensemble confidence is fragile.
- Together they explain *who* struggles and *why*, beyond RF metrics.



## Random Forests — Many Noisy Trees, One Stable Voice (Breiman, 2001)

- Train trees on bootstrapped samples with random feature subsets to decorrelate their votes.
- Aggregate those votes by majority (classification) or mean (regression) to cut variance.
- **Margin**: gap between the correct class and the runner-up; **entropy**: dispersion of votes.
- Reading the two together exposes how confident—or conflicted—the forest is, especially once aligned with  $\delta$ .

## Random Forest Margins — How Confident Is the Crowd?

$$\text{margin}(x_i) = P_{\text{correct}}(x_i) - \max_{j \neq \text{true}} P_j(x_i)$$

The **margin** measures how far ahead the correct class is over its nearest competitor.

- **High margin:** trees vote strongly for the right class → confident.
- **Low or negative margin:** trees disagree or favor another class → uncertain.

Think of it as the *vote gap* in an election — the wider the gap, the clearer the win.

# Ensemble Entropy — How Much Do Trees Disagree?

$$H(x_i) = - \sum_j P_j(x_i) \log_2 P_j(x_i)$$

The **entropy** measures how dispersed the votes are across classes.

- **Low entropy:** trees nearly unanimous → decisive prediction.
- **High entropy:** votes spread out → uncertainty or class confusion.

Within trees, entropy drives splits (purity).

Across trees, entropy reveals disagreement — the forest's collective uncertainty.

# GenAI in the Loop Scientific Experimentation

- Recursive prompting (akin to [context engineering](#)) keeps each iteration scoped.
- Ground every cycle in the `README.md` spec—goals, datasets, diagnostics.
- Automate the CLI so runs regenerate figures and tables straight into the deck.
- Commit, push, repeat: [github.com/ascott02/IRTForests](https://github.com/ascott02/IRTForests)

Plastic tubes and pots and pans

Bits and pieces and the magic from the hand - Oingo Boingo, "Weird Science" 1985

# Pipeline Overview

## Data preparation for three studies

1. Stratified CIFAR-10 subset: 10k / 2k / 2k splits. Resize 64×64, normalize, PCA → 128-D embeddings.
2. Stratified CIFAR-10 subset: 10k / 2k / 2k splits. Resize 64×64, normalize, MobileNet → 960-D embeddings.
3. MNIST mini: 4k / 800 / 800 digits, normalized 28×28 grayscale. Raw pixels.

## Random forest training

- RF (2000 trees) trained for every study; metrics and importances saved.
- Response matrices saved: CIFAR (2000 × 2000) for PCA & MobileNet, MNIST (2000 × 800).

## IRT analysis

- 1PL Rasch (SVI, 600 epochs) complete for CIFAR+PCA, CIFAR+MobileNet, and MNIST.
- 2PL (SVI, 800 epochs) complete for CIFAR+PCA, CIFAR+MobileNet, and MNIST.
- 3PL (SVI, 1000 epochs) CIFAR MobileNet only.

## Datasets Overview

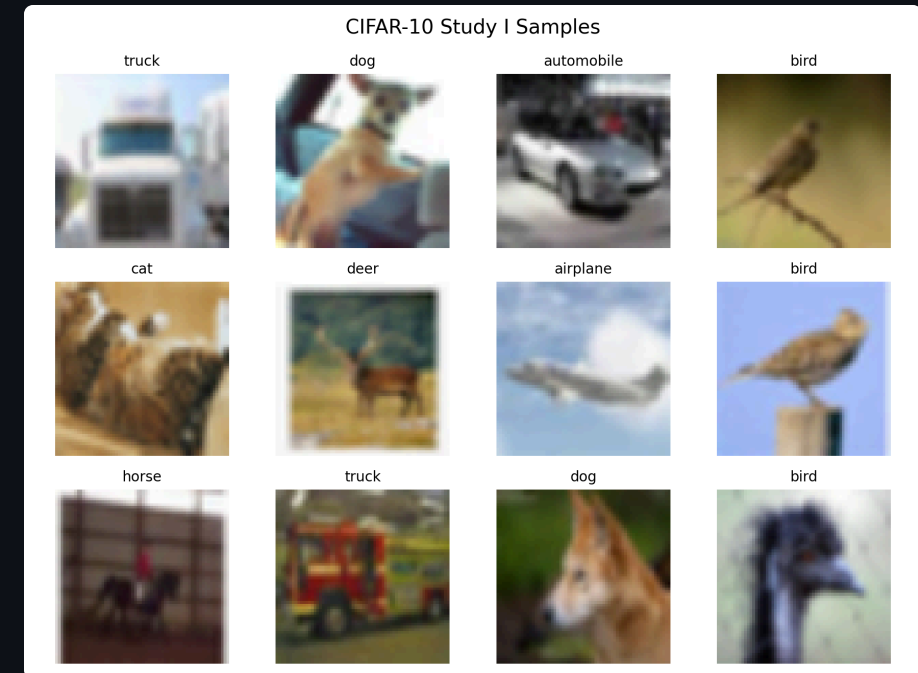
| Dataset         | Train  | Val   | Test  | Feature Pipeline                      | Notes                         |
|-----------------|--------|-------|-------|---------------------------------------|-------------------------------|
| CIFAR-10 subset | 10,000 | 2,000 | 2,000 | PCA-128 / MobileNet-V3 (960-D)        | Shared splits Study I & II    |
| MNIST mini      | 4,000  | 800   | 800   | 28×28 grayscale → raw pixels (no PCA) | Control for clean handwriting |

- CIFAR runs differ only by embeddings; labels and splits stay fixed.
- MNIST mirrors the workflow to confirm signals on cleaner data.

# Study I: CIFAR-10 + PCA-128 Embeddings

# Study I Setup: CIFAR-10 + PCA-128

- Establish the PCA baseline and capture RF uncertainty signals.
- Use IRT to pinpoint weak trees and hard items that motivate stronger features.
- Fix a stratified CIFAR-10 split (10k / 2k / 2k).
- Train 2000 trees and score them on the shared test set.
- Build a  $2000 \times 2000$  response matrix (mean tree accuracy  $\approx 0.18$ ).
- Artifacts: metrics, margins, entropy, IRT outputs.



Study I sample grid — stratified CIFAR-10 slices

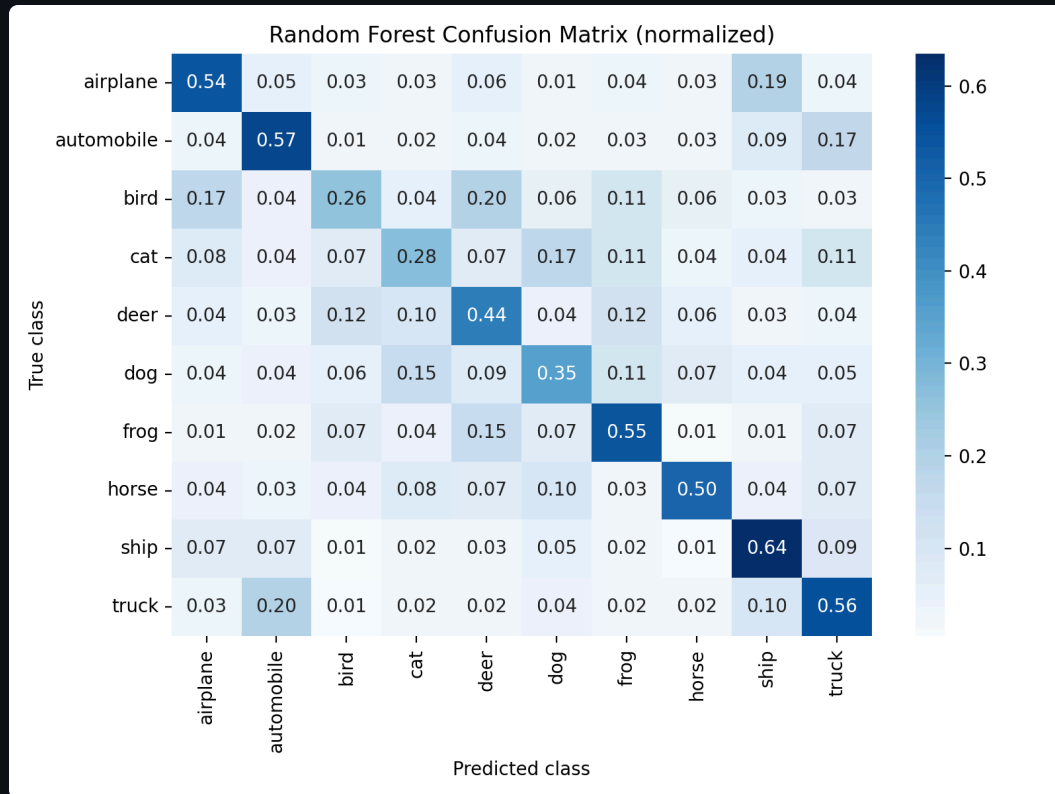


## Study I Performance (PCA-128)

| Metric                                                | Value                       |
|-------------------------------------------------------|-----------------------------|
| Test / Val / OOB acc                                  | 0.468 / 0.470 / 0.442       |
| Per-class range                                       | 0.260 (bird) → 0.635 (ship) |
| Mean tree accuracy                                    | 0.1763                      |
| Mean margin / entropy                                 | 0.0058 / 2.1723             |
| $\delta$ negatively correlates with margin (Pearson)  | -0.815                      |
| $\delta$ positively correlates with entropy (Pearson) | 0.687                       |

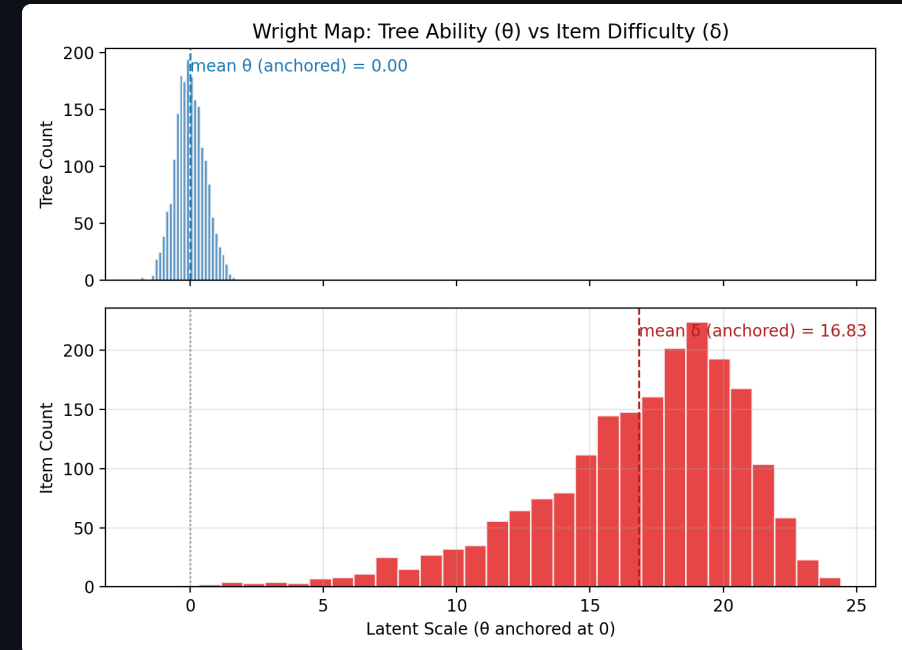
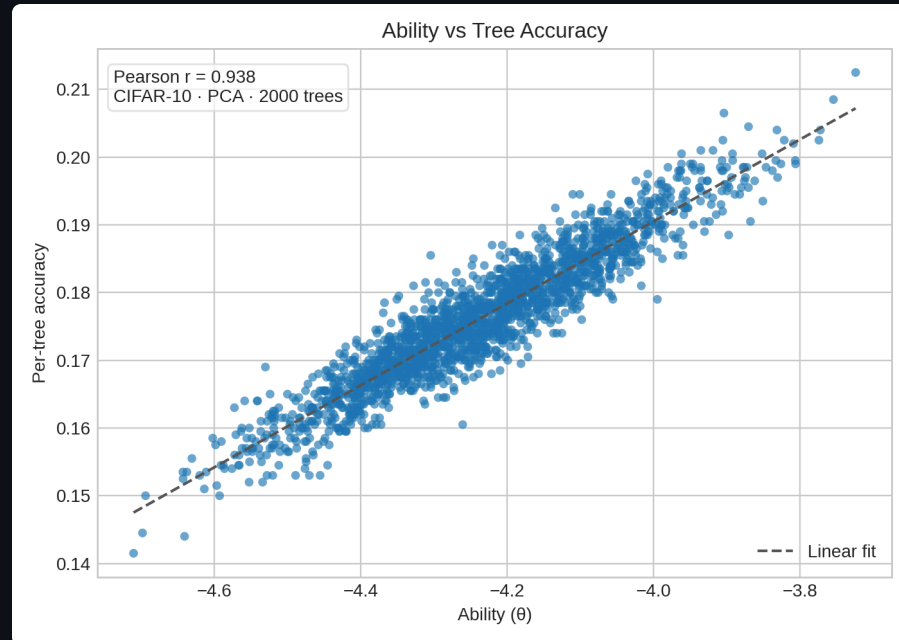
- Baseline ensemble still underperforms due to weak PCA features yet preserves  $\delta$  alignment.
- Margins hover near zero (mean  $\approx 0.006$ ) and entropy stays high (2.17), signalling broad disagreement—prime for IRT.

# Study I Confusion Matrix



- Off-diagonal spikes (cat vs dog, bird vs airplane, horse vs deer) mirror high- $\delta$  items.
- Ships and trucks still lead the diagonal ( $\approx 64\%$  /  $56\%$  accuracy), yet well short of a clean block —further underscoring the curation need.

# Study I Diagnostics: Ability Profiles

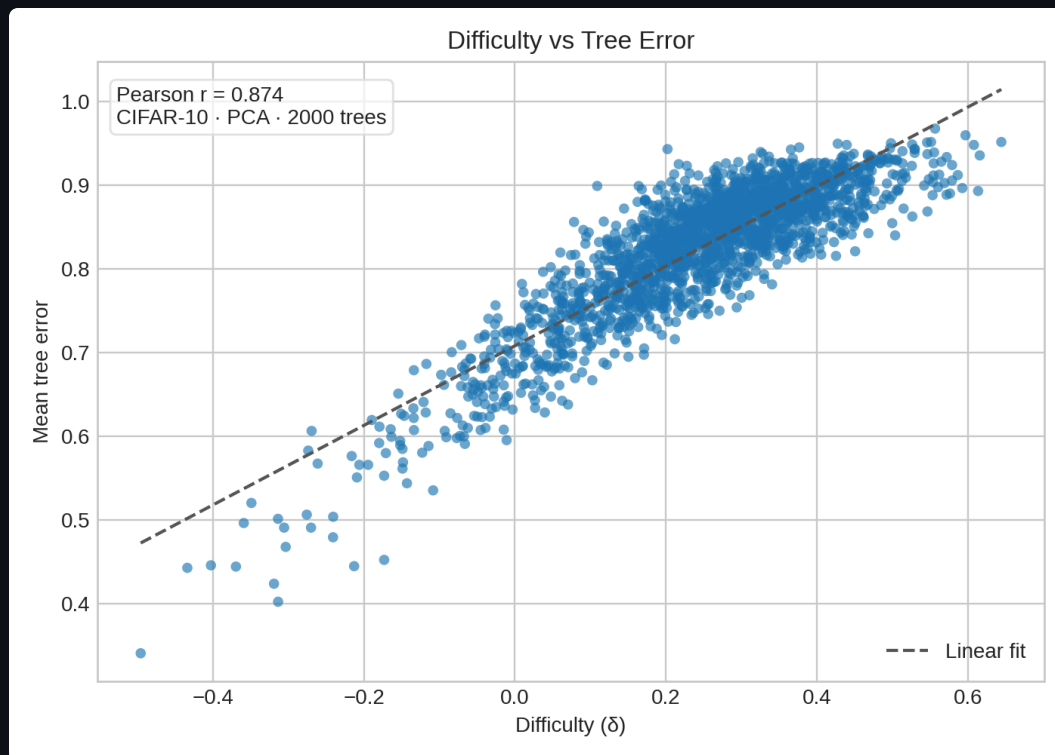


Ability ( $\theta$ ) vs tree accuracy — Spearman  $\approx 0.99$

Wright map:  $\delta$  pushes deep into the positive tail

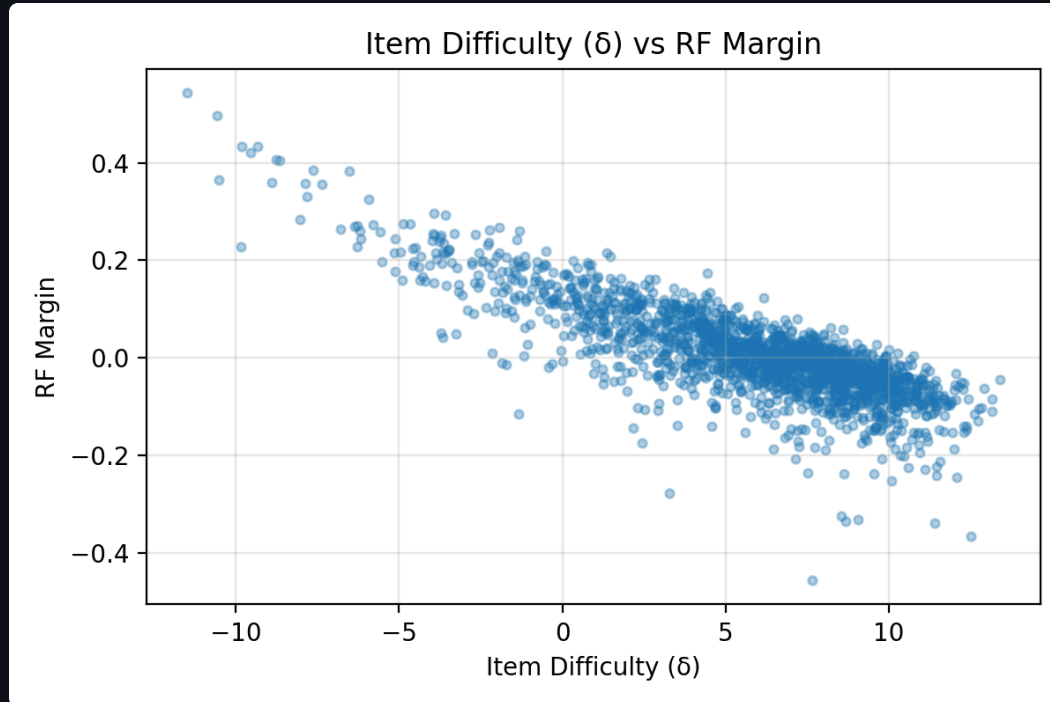
- $\theta$  falls between roughly  $-1.9$  and  $+2.1$  logits, so even modest shifts still separate stronger trees by a few percentage points.
- $\delta$  ranges from about  $-0.5$  to  $+24$  logits after centering, highlighting how ambiguous animal items sit far from the easy tail.

# Study I Diagnostics: $\delta$ vs Error Rate

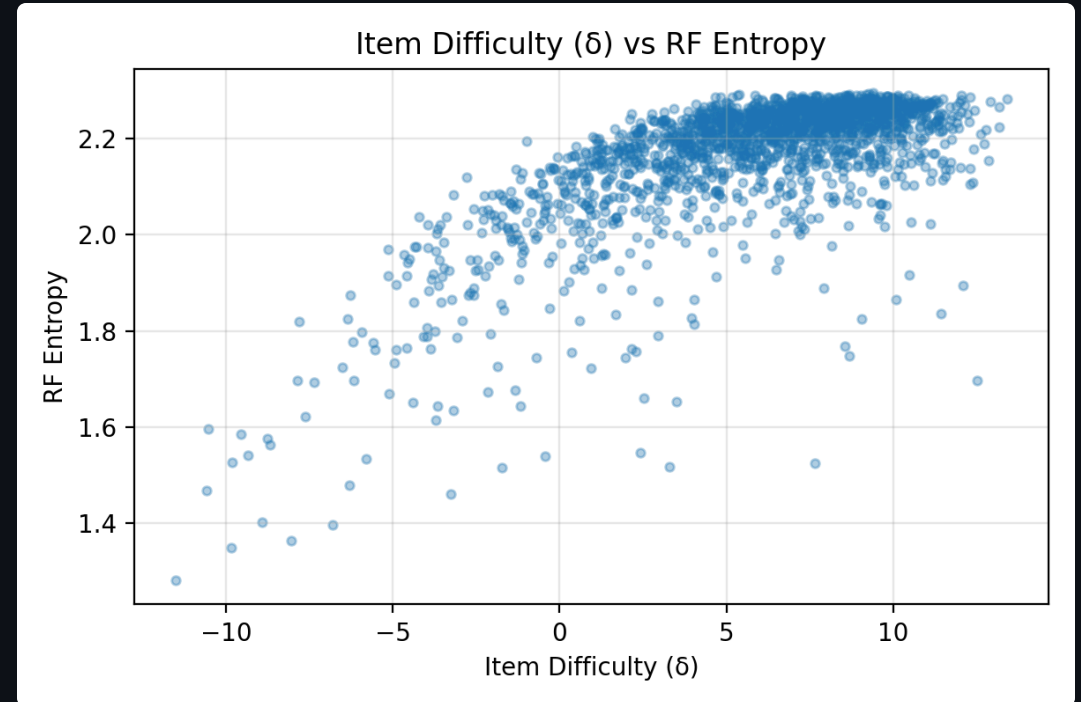


- $\delta > 0.4$  maps to >80% tree error—mostly ambiguous animals — while  $\delta < -0.3$  are trivially easy.
- Pearson  $\approx 0.87$ , Spearman  $\approx 0.82$ : difficulty doubles as an error heat-map.

# Study I Diagnostics: $\delta$ vs RF Signals



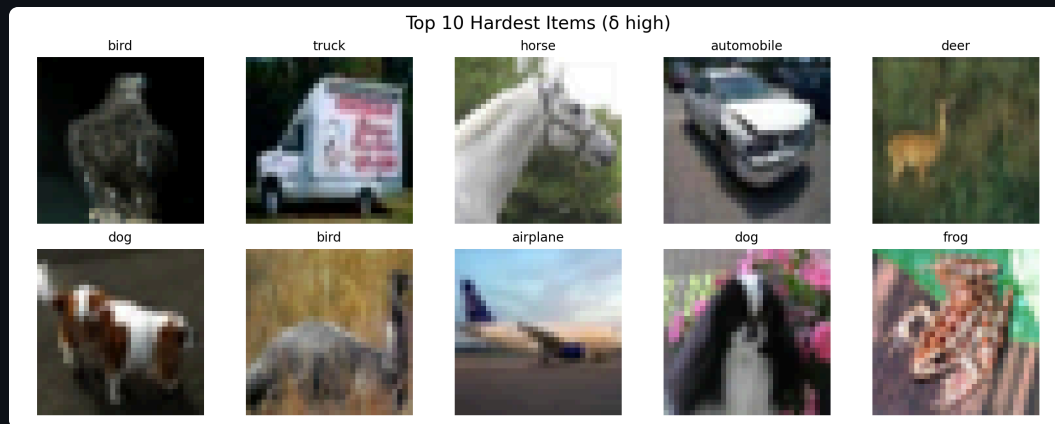
PCA run:  $\delta$  vs margin (Pearson  $-0.82$ )



PCA run:  $\delta$  vs entropy (Pearson  $0.69$ )

- Hard items cluster bottom-right (low margin, high entropy); opposite corner houses easy wins.
- Study II mirrors the trend with even stronger correlations.

# Study I Evidence: Hard vs Easy Examples



- Hardest items skew toward ambiguous airplane/ship silhouettes and cluttered cat/dog scenes.
- Easy set is dominated by high-contrast cues (e.g., red fire trucks), yielding low  $\delta$  and entropy.

# Study I Fit Checks & Edge Cases

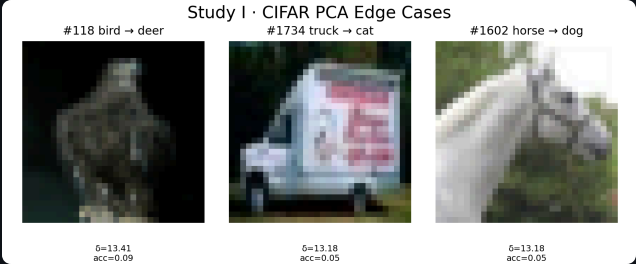
## Fit diagnostics

| Metric                  | Value       |
|-------------------------|-------------|
| Item infit $\mu$ / p95  | 0.18 / 0.35 |
| Item outfit $\mu$ / p95 | 0.18 / 0.34 |
| Tree infit $\mu$ / p95  | 0.35 / 0.48 |
| Tree outfit $\mu$ / p95 | 0.18 / 0.19 |

- MSQs well below 1 show tree responses are steadier than a pure Rasch prior;  $|z|$  never exceeds 0.05.

## Edge cases worth a look

- #118 bird  $\rightarrow$  deer votes ( $\delta \approx 13.4$ , margin  $\approx -0.05$ , entropy  $\approx 2.28$ ).
- #1734 truck  $\rightarrow$  cat/frog split ( $\delta \approx 13.2$ , margin  $\approx -0.09$ , entropy  $\approx 2.27$ ).
- #1602 horse  $\rightarrow$  dog/horse tie ( $\delta \approx 13.2$ , margin  $\approx -0.11$ , entropy  $\approx 2.22$ ).
- Each item sits below 9% tree accuracy—prime targets for relabeling or curated augmentations.



Study I edge cases · IDs 118, 1734, 1602

## Study I Takeaways

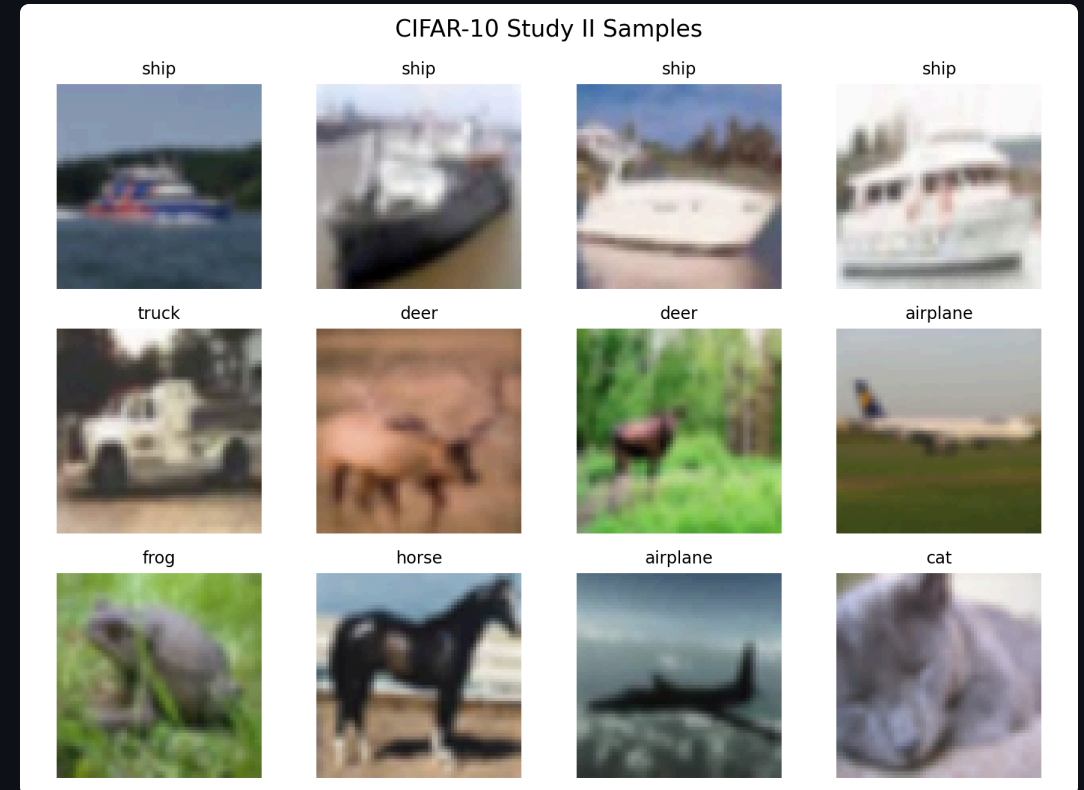
- Weak PCA features create long tails in both ability ( $\theta$ ) and difficulty ( $\delta$ ), exposing erratic trees.
- Margin and entropy correlate with  $\delta$ , but clusters of high-difficulty animals persist across diagnostics.
- Visual inspection confirms mislabeled or low-signal items driving high  $\delta$ , motivating feature upgrades.



## Study II: CIFAR-10 + MobileNet Embeddings

# Study II Setup: CIFAR-10 + MobileNet-V3

- Hold the splits fixed to isolate feature gains.
- Swap PCA features for MobileNet-V3 (960-D) while keeping tree count and splits constant.
- Test whether richer embeddings tighten  $\theta$  spread and retain  $\delta$  alignment.
- Compare RF metrics, uncertainty signals, and IRT parameters against the baseline.



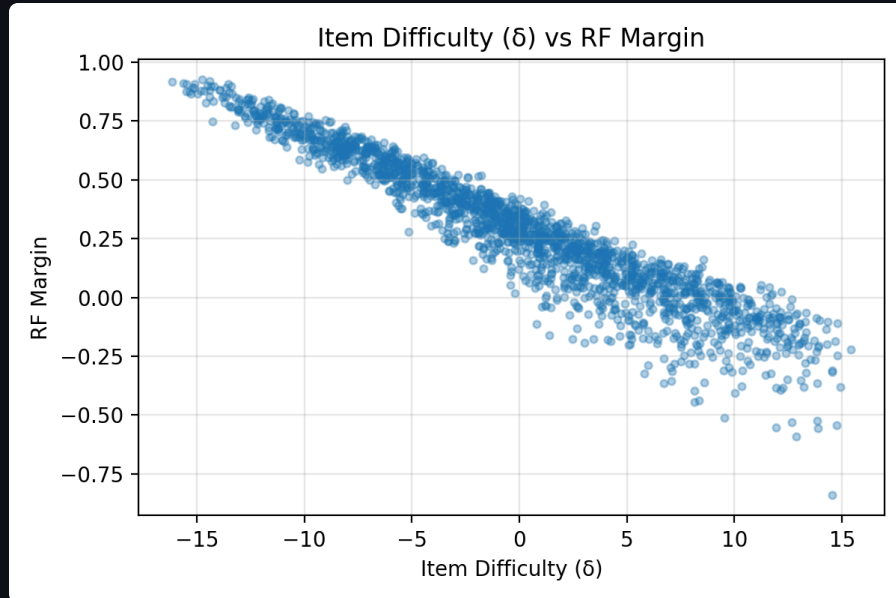
Study II sample grid — same splits, MobileNet embeddings

## Study II Performance (MobileNet-V3)

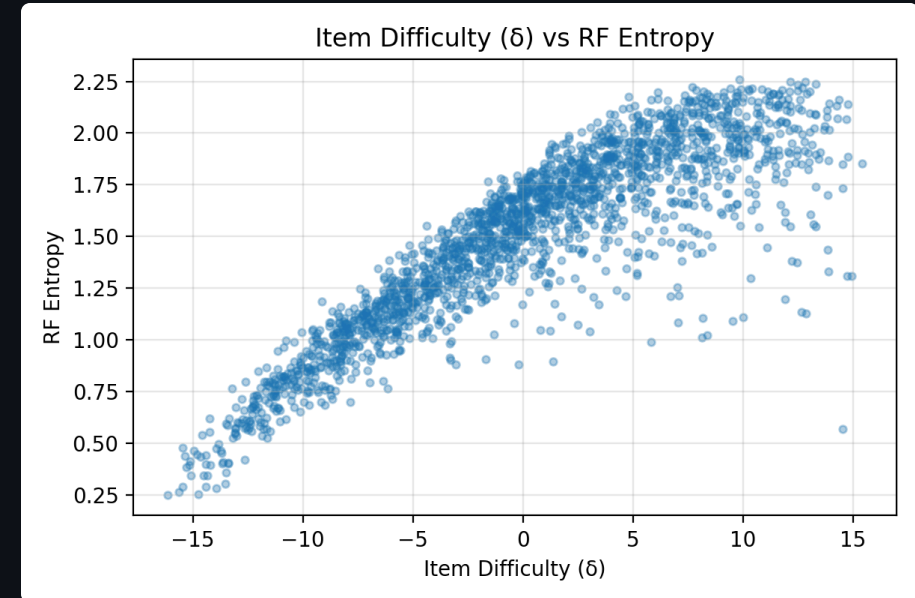
| Metric                                                | Value                       |
|-------------------------------------------------------|-----------------------------|
| Test / Val / OOB acc                                  | 0.819 / 0.820 / 0.812       |
| Per-class range                                       | 0.695 (bird) → 0.925 (ship) |
| Mean tree accuracy                                    | 0.4792                      |
| Mean margin / entropy                                 | 0.2806 / 1.4929             |
| $\delta$ negatively correlates with margin (Pearson)  | -0.950                      |
| $\delta$ positively correlates with entropy (Pearson) | 0.881                       |

- Pretrained features boost accuracy by 35 pp while strengthening  $\delta$  correlations.
- Higher margins and lower entropy show confidence gains except on stubborn animal classes.
- Artifacts: metrics, response matrix, signals, and IRT outputs under `data/mobilenet/`.

# Study II Diagnostics: $\delta$ vs RF Signals



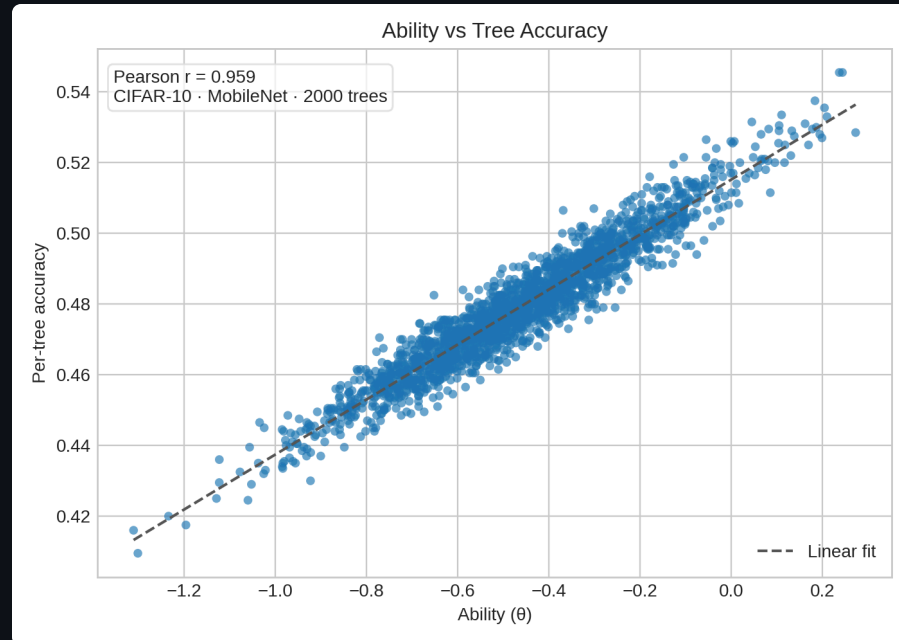
$\delta$  vs margin (Pearson  $-0.95$ )



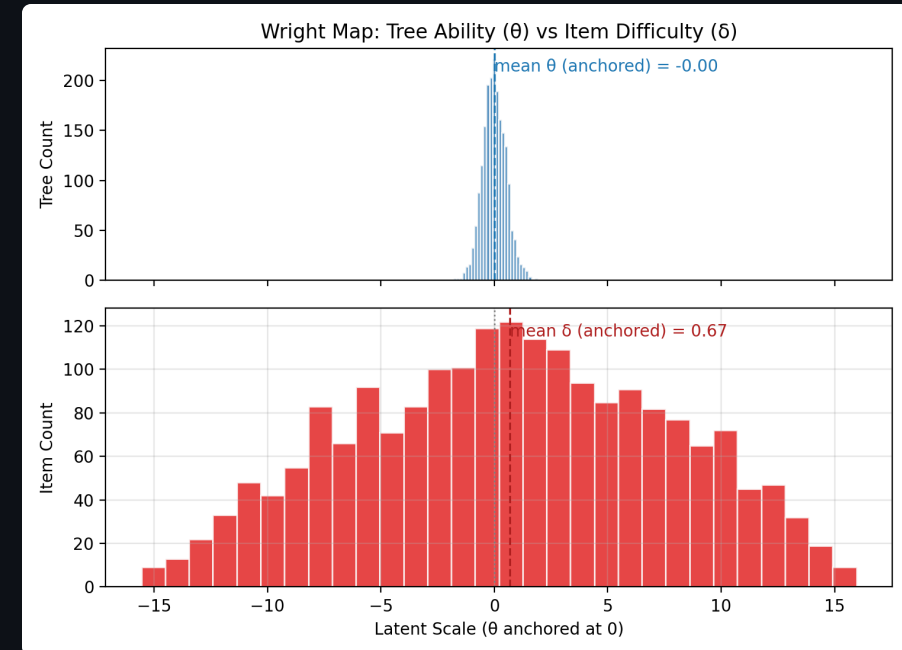
$\delta$  vs entropy (Pearson  $0.88$ )

- MobileNet compresses the easy cluster (high margin, low entropy) while isolating true hard cases.
- Larger  $|\text{corr}|$  values show tighter agreement between  $\delta$  and RF uncertainty.
- Cat/dog confusions persist, marking curation targets.

# Study II Diagnostics: Ability Profiles



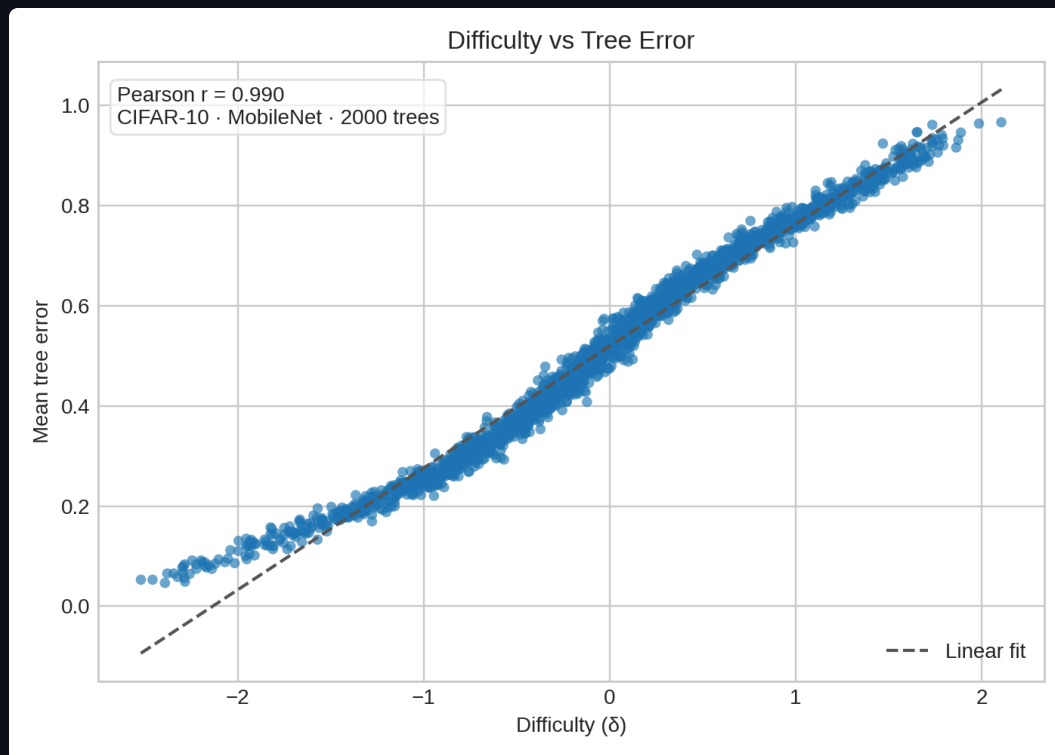
Ability ( $\theta$ ) vs tree accuracy — Pearson 0.96



Wright map:  $\delta$  stretches about  $\pm 16$  logits

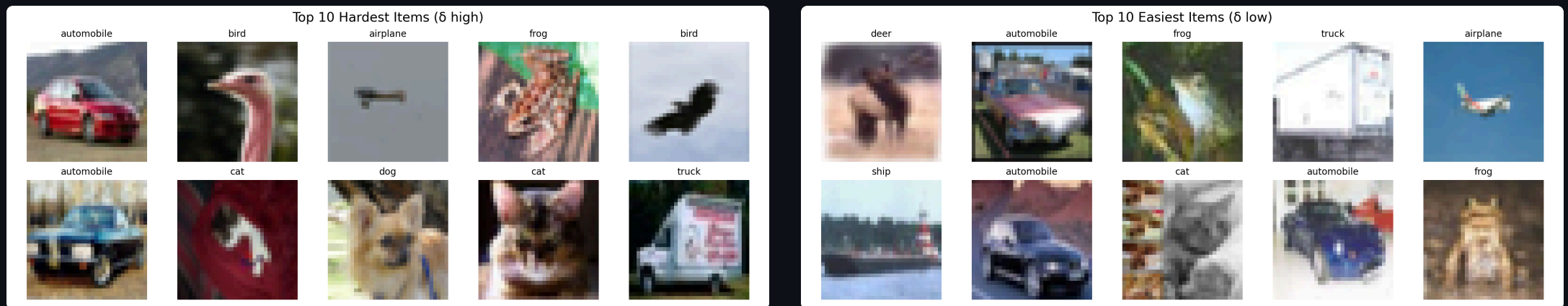
- $\theta$  stays within roughly  $\pm 2$  logits, keeping the ensemble tightly banded while still ranking trees cleanly.
- Ability remains tied to per-tree accuracy, so feature quality—rather than tree diversity—now caps gains.

## Study II Diagnostics: $\delta$ vs Error Rate



- Pearson 0.99 keeps  $\delta$  aligned with mean tree error even at the higher accuracy ceiling.
- Hardest items ( $\delta > 1.5$ ) persist—mostly cat/dog overlaps and ambiguous aircraft—while the easy zone ( $\delta < -1$ ) expands.

# Study II Evidence: Hard vs Easy Examples



- MobileNet tightens easy clusters yet the same cat/dog outliers survive with  $\delta > 1.5$ .
- Easy wins sharpen into high-contrast ships and trucks, showing how feature upgrades cleanly separate low- $\delta$  items.

# Study II Fit Checks & Edge Cases

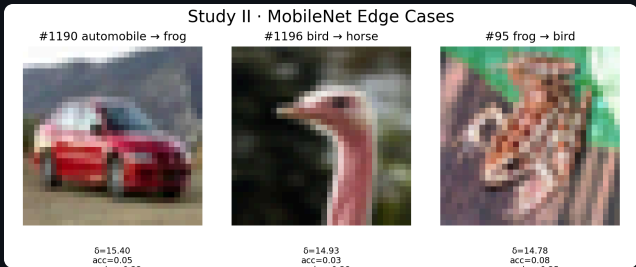
## Fit diagnostics

| Metric                  | Value       |
|-------------------------|-------------|
| Item infit $\mu$ / p95  | 0.27 / 0.37 |
| Item outfit $\mu$ / p95 | 0.27 / 0.37 |
| Tree infit $\mu$ / p95  | 0.29 / 0.31 |
| Tree outfit $\mu$ / p95 | 0.27 / 0.29 |

- Narrow MSQ spread ( $\leq 0.37$ ) confirms MobileNet trees behave consistently; no misfit flags at  $|z| > 0.05$ .

## Edge cases worth a look

- #1190 automobile  $\rightarrow$  frog votes ( $\delta \approx 15.4$ , margin  $\approx -0.22$ , entropy  $\approx 1.85$ ; top probs frog 0.28, deer 0.27).
- #1196 bird  $\rightarrow$  horse ( $\delta \approx 14.9$ , margin  $\approx -0.38$ , entropy  $\approx 1.31$ ; horse 0.41, deer 0.41, bird 0.08).
- #95 frog  $\rightarrow$  bird ( $\delta \approx 14.8$ , margin  $\approx -0.25$ , entropy  $\approx 1.89$ ; bird 0.32, deer 0.20, frog 0.17).
- These persistent outliers survive the feature upgrade—queue them for image/label review next.





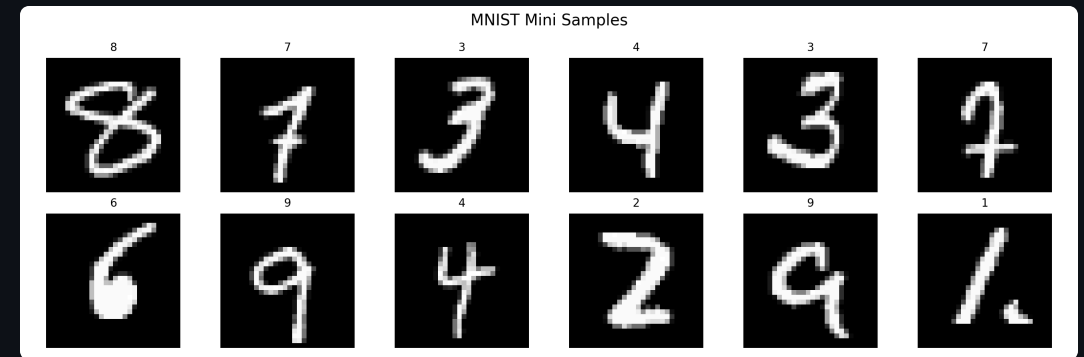
## Study II Takeaways

- MobileNet embeddings add 35 pp of accuracy while maintaining a focused ability band ( $\text{Std}(\theta) \approx 0.23$ ).
- $\delta$  stays aligned with RF uncertainty, isolating a smaller yet stubborn ambiguous cluster.
- Residual cat/dog confusion points to data curation as the next lever.

## Section III · Control Study (MNIST)

## Study III Setup: MNIST Mini-Study

- Probe the pipeline on a high-signal, low-noise dataset.
- Use a lightweight handwriting set to validate RF  $\times$  IRT beyond CIFAR-10.
- Confirm that IRT still mirrors RF uncertainty when accuracy is near perfect.
- Treat it as a control case where ambiguity is rare yet still detectable.



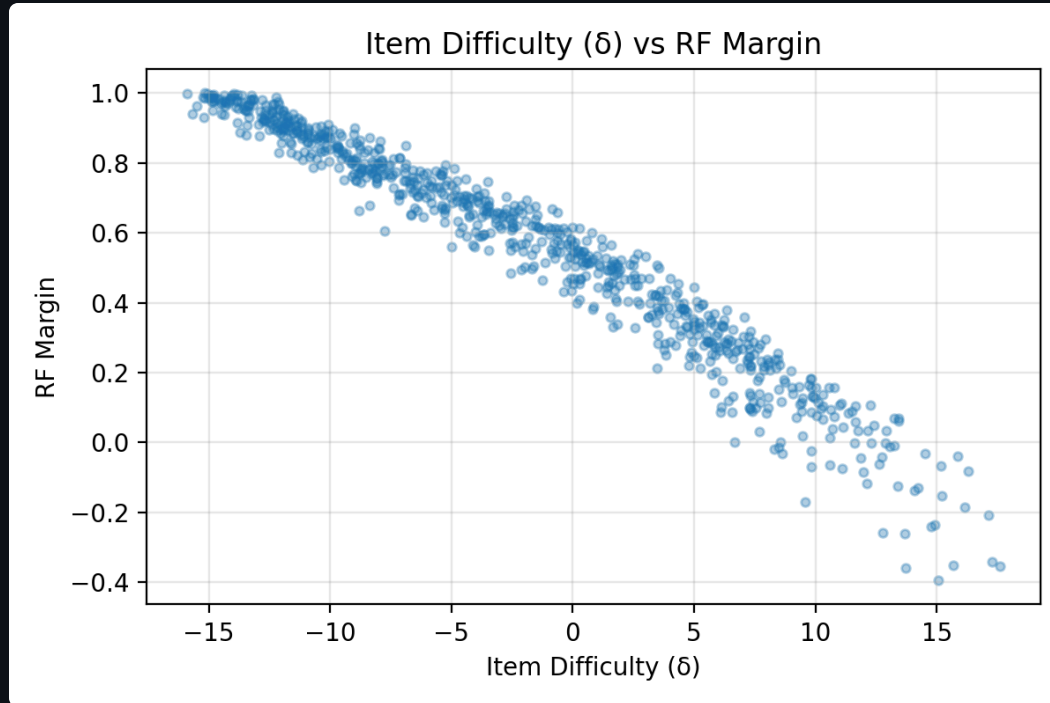
Study III sample grid — curated MNIST mini split

## Study III Performance (MNIST)

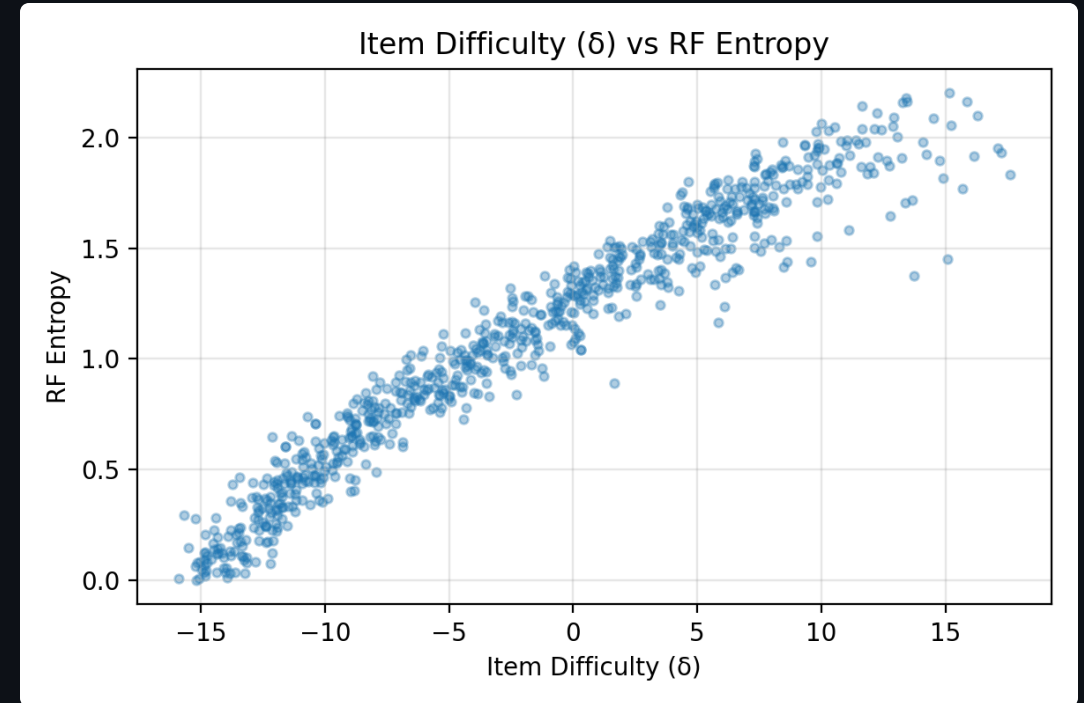
| Metric                                                | Value                 |
|-------------------------------------------------------|-----------------------|
| Train / Val / Test                                    | 4000 / 800 / 800      |
| RF test / val / OOB                                   | 0.954 / 0.944 / 0.939 |
| Mean margin / entropy                                 | 0.5644 / 1.0768       |
| $\delta$ negatively correlates with margin (Pearson)  | -0.975                |
| $\delta$ positively correlates with entropy (Pearson) | 0.970                 |
| $\theta$ mean $\pm$ std                               | 3.04 $\pm$ 0.29       |
| $\delta$ mean $\pm$ std                               | -0.13 $\pm$ 0.47      |

- Ambiguous digits (e.g., brushed 5 vs 6) still spike  $\delta$  toward the positive tail; elsewhere the forest is decisive.
- Low entropy + high margin line up with low  $\delta$ , giving a “sanity benchmark” beyond CIFAR.

# Study III Diagnostics: $\delta$ vs RF Signals



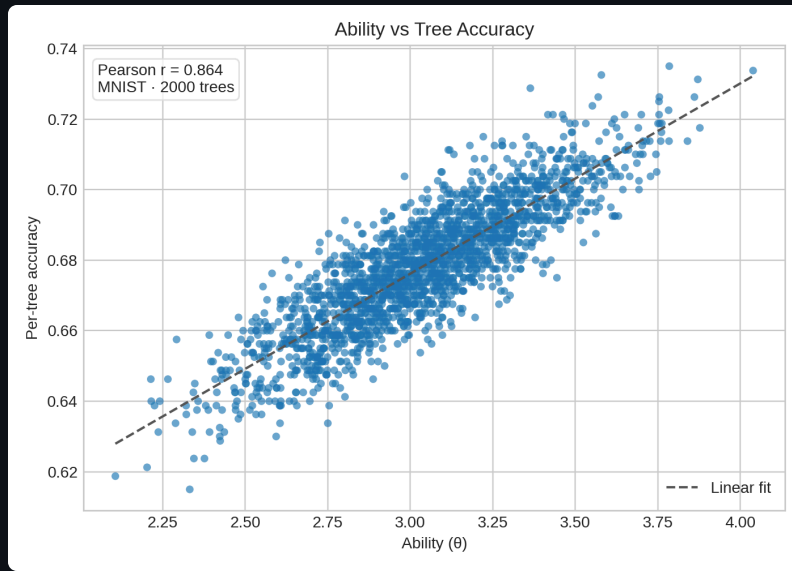
$\delta$  vs margin (Pearson  $-0.97$ )



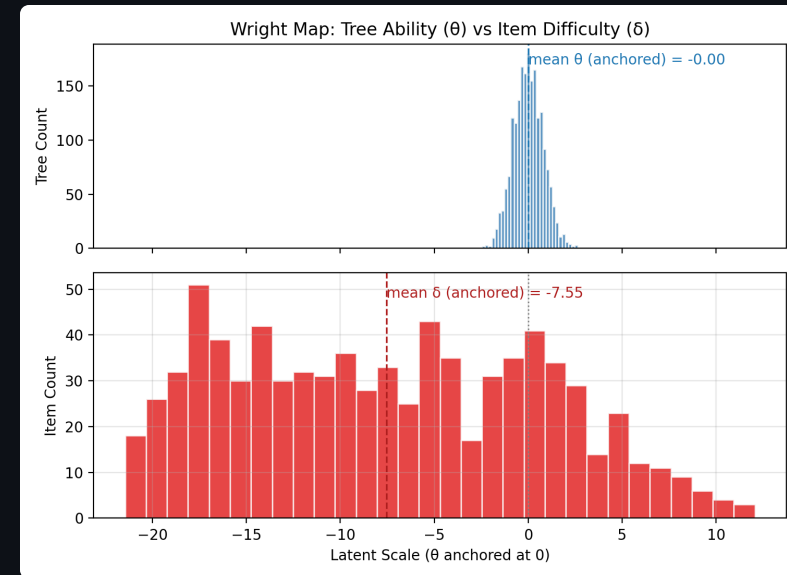
$\delta$  vs entropy (Pearson  $0.97$ )

- Clean digits show near-perfect alignment between  $\delta$  and RF uncertainty.
- Only a handful of  $\delta > 1.2$  digits drive the residual uncertainty (stroke collisions like 3/5, 4/9).

# Study III Diagnostics: Ability Profiles



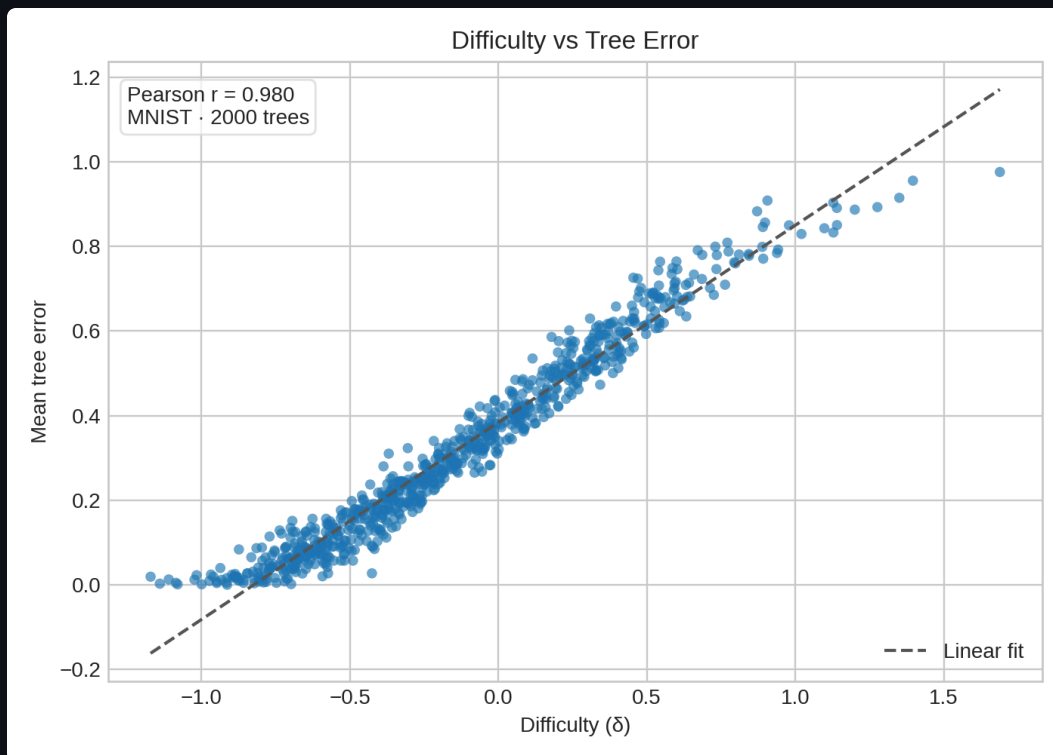
Ability ( $\theta$ ) vs tree accuracy — Pearson 0.98



Wright map: most digits land several logits below

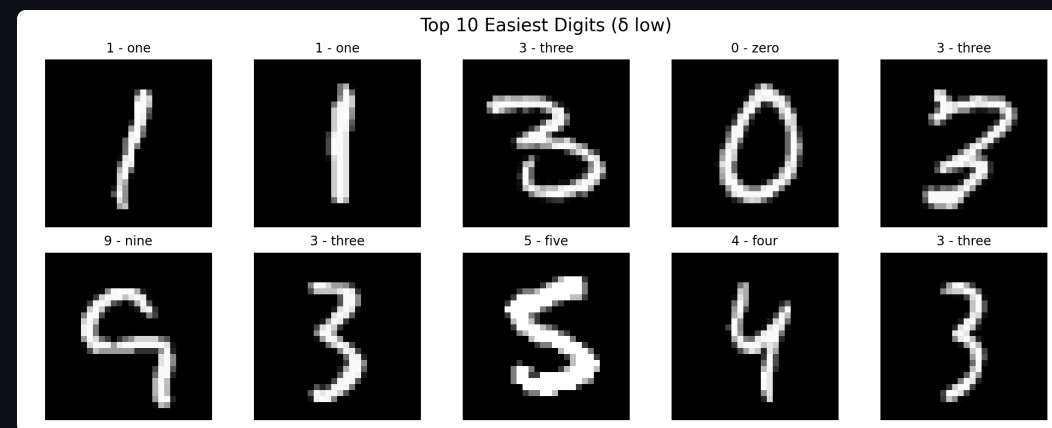
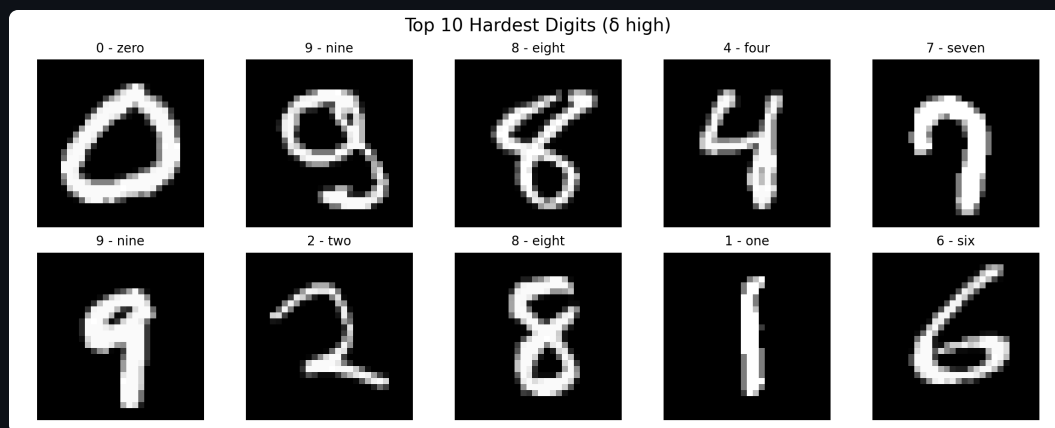
- $\theta$  covers roughly  $-2.5$  to  $+2.6$  logits, underscoring strong consensus, while most  $\delta$  values sit several logits below the trees with only a handful near the origin.
- Shared scales expose plentiful easy wins with only a few sharp spikes—opposite of the CIFAR baseline.

## Study III Diagnostics: $\delta$ vs Error Rate



- Pearson 0.98 keeps  $\delta$  tied to mean tree error despite the high accuracy ceiling.
- $\delta > 1.2$  corresponds to stroke-collided 3/5/8 and 4/9 pairs; the long negative tail is trivial for the ensemble.

## Study III Evidence: Hard vs Easy Digits



- Hardest digits show stroke collisions (3 vs 5, 4 vs 9) that push  $\delta$  above 1 despite high margins elsewhere.
- Easy digits are crisp, centered strokes—useful anchors when explaining why  $\delta$  plunges on most of the dataset.



# Study III Fit Checks & Edge Cases

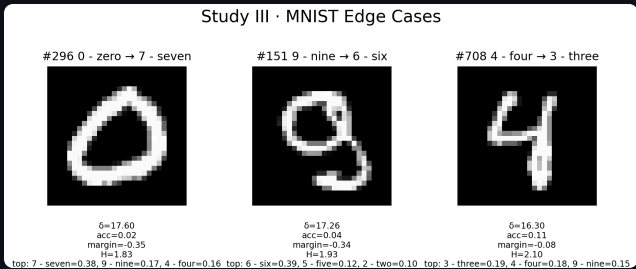
## Fit diagnostics

| Metric                  | Value       |
|-------------------------|-------------|
| Item infit $\mu$ / p95  | 0.23 / 0.38 |
| Item outfit $\mu$ / p95 | 0.22 / 0.37 |
| Tree infit $\mu$ / p95  | 0.30 / 0.32 |
| Tree outfit $\mu$ / p95 | 0.22 / 0.25 |

- Rasch residuals stay tight ( $|z| < 0.07$ ), confirming the control study's consistency.

## Edge cases worth a look

- #296 digit 0  $\rightarrow$  vote 7 ( $\delta \approx 17.6$ , margin  $\approx -0.35$ , entropy  $\approx 1.83$ ; top probs 7=0.38, 9=0.18, 4=0.16).
- #151 digit 9  $\rightarrow$  vote 6 ( $\delta \approx 17.3$ , margin  $\approx -0.34$ , entropy  $\approx 1.93$ ; top probs 6=0.39, 5=0.12, 2=0.10).
- #708 digit 4  $\rightarrow$  vote 3 ( $\delta \approx 16.3$ , margin  $\approx -0.08$ , entropy  $\approx 2.10$ ; top probs 3=0.19, 4=0.18, 9=0.15).
- Archive these strokes for a “confusing digits” gallery or curation playbook.



## Study III Takeaways

- $\delta$  and RF uncertainty agree almost perfectly, while  $\theta$  stays high yet still flags the rare ambiguous strokes.
- The control study confirms the RF  $\times$  IRT pipeline holds outside noisy vision data.

## Section IV · Cross-Study & Diagnostics

- Compare backbones and datasets on a shared  $\theta/\delta$  scale.
- Surface recurring themes before the close.

# Cross-Study Snapshot

| Study                       | Feature Backbone     | Test Acc | $\delta$ negatively correlates with margin (Pearson) | $\delta$ positively correlates with entropy (Pearson) | Std( $\theta$ ) | Std( $\delta$ ) |
|-----------------------------|----------------------|----------|------------------------------------------------------|-------------------------------------------------------|-----------------|-----------------|
| Study I: CIFAR + PCA-128    | PCA-128              | 0.468    | -0.815                                               | 0.687                                                 | 0.154           | 0.150           |
| Study II: CIFAR + MobileNet | MobileNet-V3 (960-D) | 0.819    | -0.950                                               | 0.881                                                 | 0.228           | 0.871           |
| Study III: MNIST Mini       | Raw pixels           | 0.954    | -0.975                                               | 0.970                                                 | 0.289           | 0.472           |

- *Std( $\theta$ ) measures tree ability spread; Std( $\delta$ ) measures item difficulty spread.*
- $\delta$  stays negative with margin and positive with entropy for every study (-0.82/-0.95/-0.98 vs +0.69/+0.88/+0.97).
- $\theta$  spread remains compact (Std( $\theta$ )  $\approx$  0.15-0.29); MobileNet is only slightly wider as headroom grows.
- Difficulty variance jumps on MobileNet (Std( $\delta$ )  $\approx$  0.87) while MNIST stays moderate, highlighting how rich features surface nuanced “hard” digits.

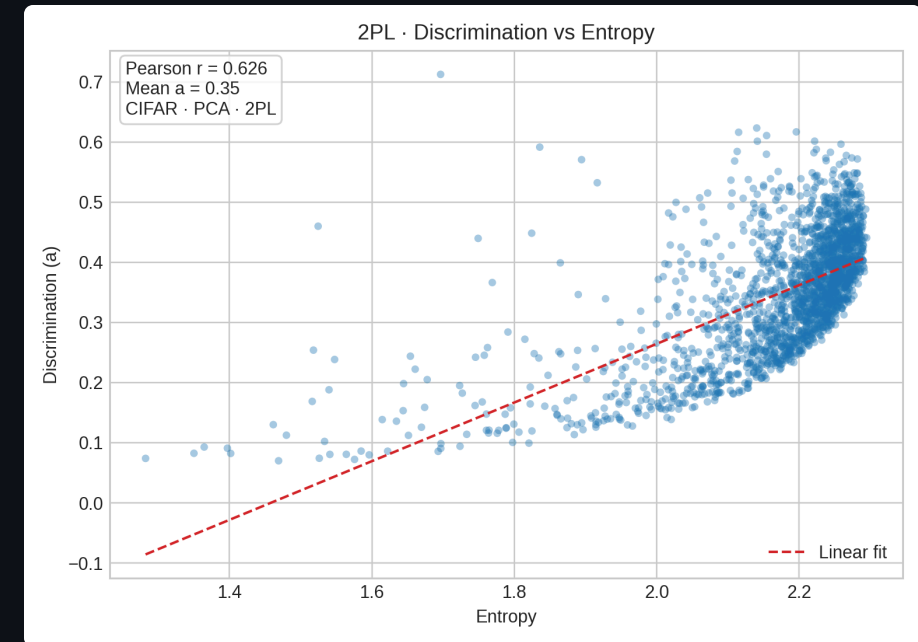
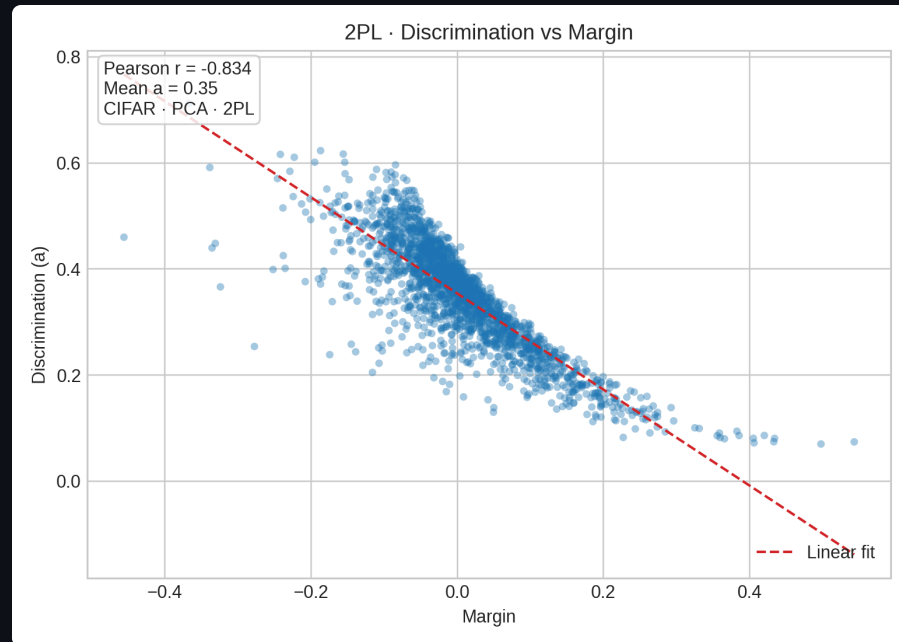
# Cross-Study Fit Snapshot

| Study             | Item infit $\mu$ / p95 | Item outfit $\mu$ / p95 | Tree infit $\mu$ / p95 | Tree outfit $\mu$ / p95 |
|-------------------|------------------------|-------------------------|------------------------|-------------------------|
| CIFAR + PCA       | 0.18 / 0.35            | 0.18 / 0.34             | 0.35 / 0.48            | 0.18 / 0.19             |
| CIFAR + MobileNet | 0.27 / 0.37            | 0.27 / 0.37             | 0.29 / 0.31            | 0.27 / 0.29             |
| MNIST mini        | 0.23 / 0.38            | 0.22 / 0.37             | 0.30 / 0.32            | 0.22 / 0.25             |

- All MSQs stay well below 1, indicating over-dispersed errors are rare and Rasch assumptions hold after 2000-tree scaling.
- MobileNet’s slight lift in item MSQ reflects richer feature diversity, while MNIST keeps both item and tree fits exceptionally tight.

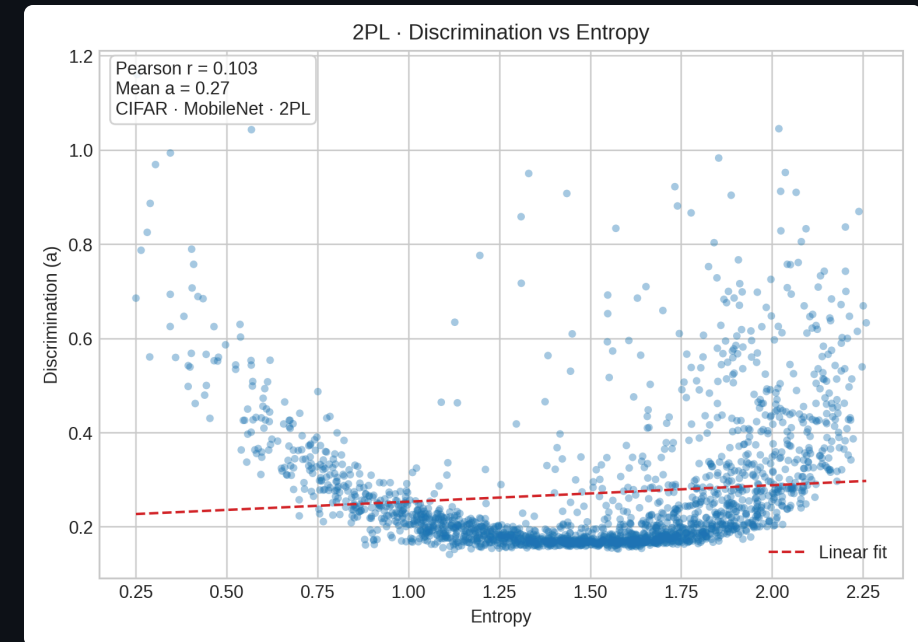
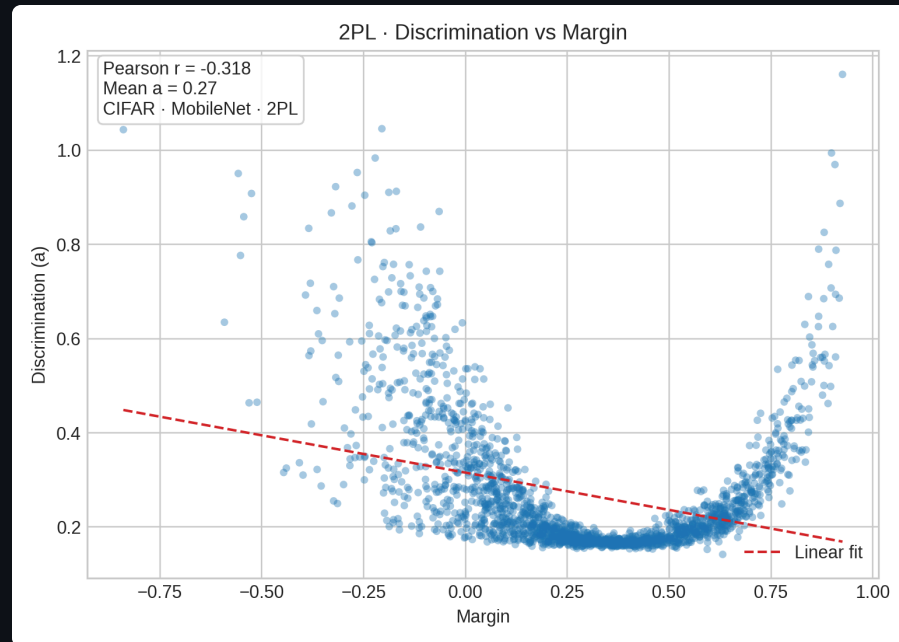
# 2PL Discrimination (CIFAR + PCA)

- 800-epoch 2PL fit (lr 0.02) yields mean  $a \approx 0.35 \pm 0.10$  (range 0.07–0.71).
- $a$  correlates with margin at  $-0.83$  and with entropy at  $+0.63$ , aligning slope with RF uncertainty signals.
- Discrimination peaks on the low-margin, high-entropy animal items and steadily tapers for easier scenes, leaving high-margin images with softer slopes.



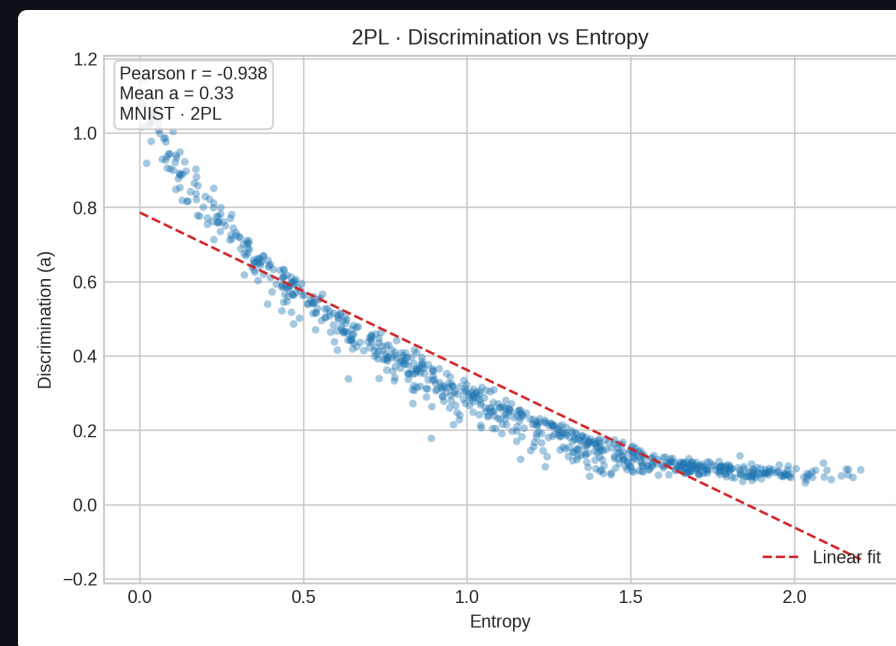
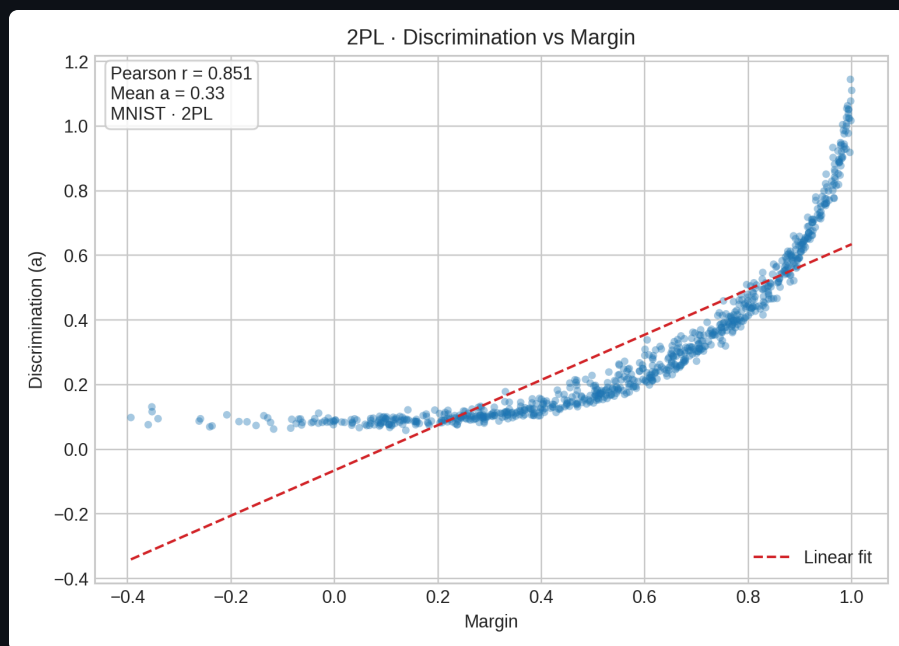
# 2PL Discrimination (CIFAR + MobileNet)

- Mean  $a$  settles at  $0.27 \pm 0.15$  with a modest tail (max  $\approx 1.16$ ).
- $a$  correlates with margin at  $-0.32$  and with entropy at  $+0.10$ , keeping residual cat/dog confusion in focus while the easy cluster sharpens.
- Discrimination concentrates in the tails: hard animal confusions and trivially easy scenes separate trees, while mid-uncertainty items contribute little.



# 2PL Discrimination (MNIST)

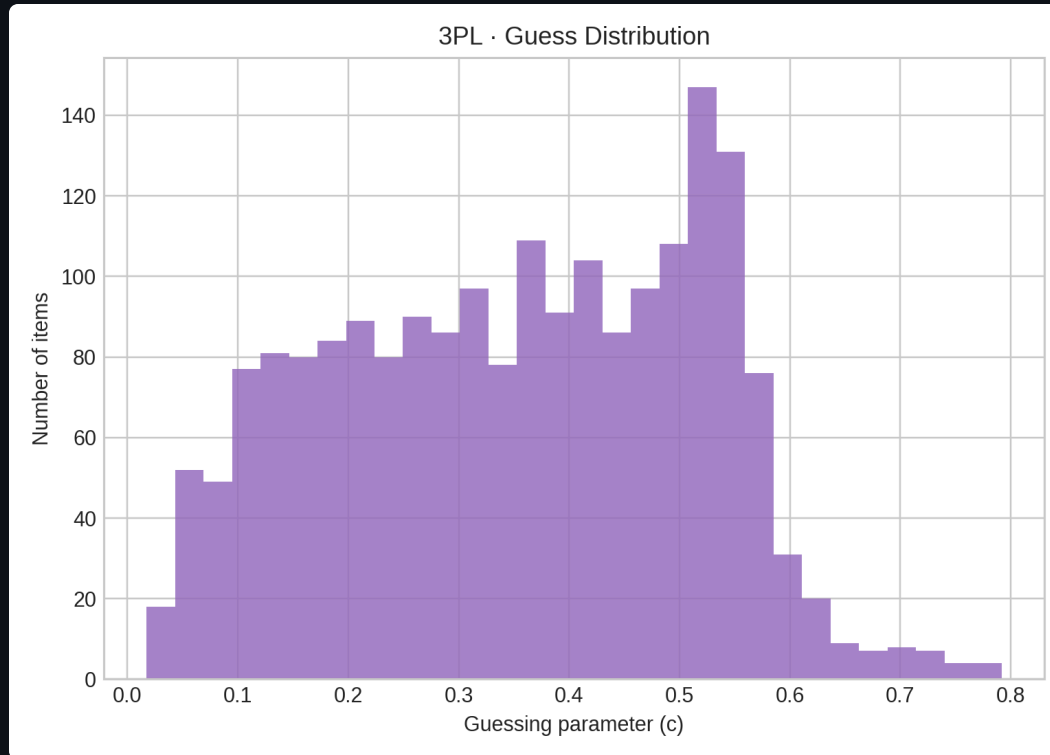
- Mean  $a$  lifts to  $0.33 \pm 0.25$ , so only a modest slice of digits remains truly separating despite the high accuracy ceiling.
- $a$  correlates with margin at  $+0.89$  while its correlation with entropy flips to  $-0.96$ —uncertainty vanishes outside the awkward strokes.
- Discrimination climbs with margin and falls with entropy: crisp, easy digits carry the steepest slopes while ambiguous stroke collisions stay much flatter.



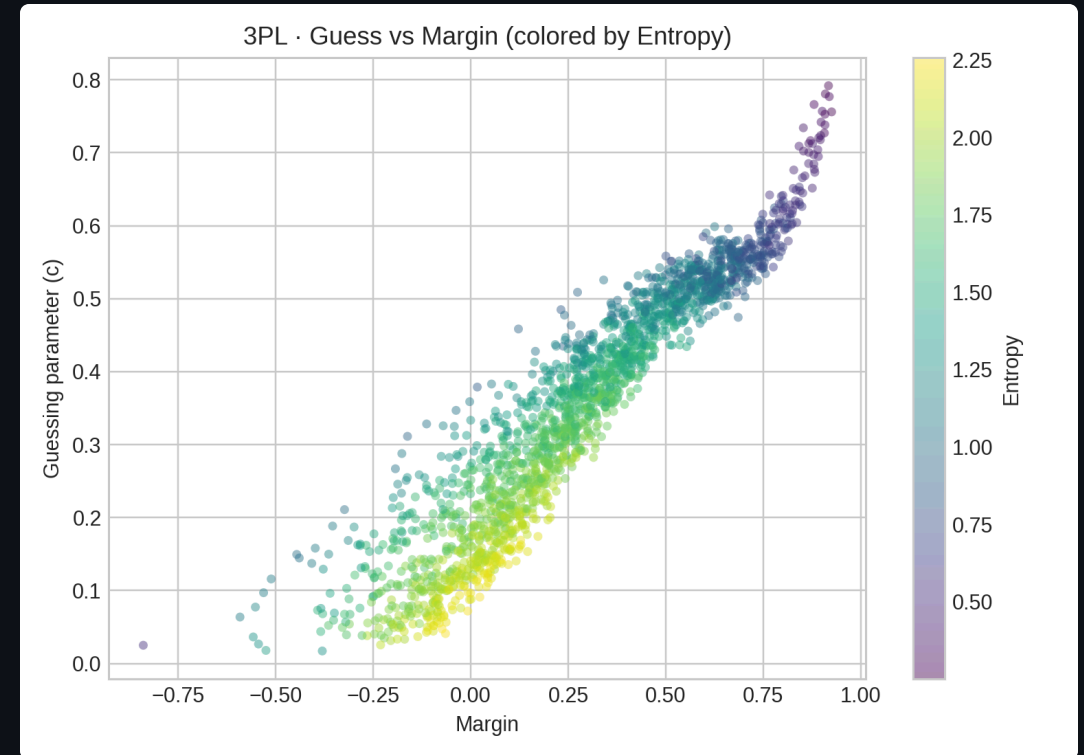


# 3PL Pilot · MobileNet

- 1k-epoch 3PL run (lr 0.01) lands at guess mean  $0.35 \pm 0.16$ .
- $\theta$  vs accuracy stays tight (Pearson 0.98); slopes average  $0.32 \pm 0.08$  with a broader tail.
- High guess mass piles onto the ambiguous animal scenes (low margin, high entropy), reinforcing the “guessing” narrative.



3PL MobileNet · Guess distribution

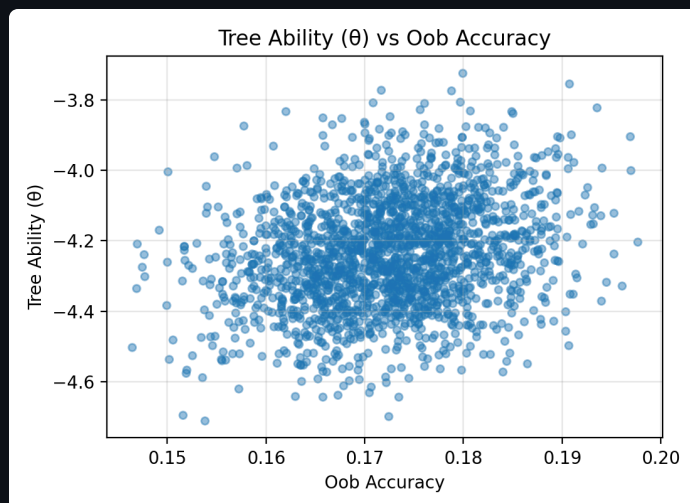


3PL MobileNet · Guess vs Margin (colored by entropy)

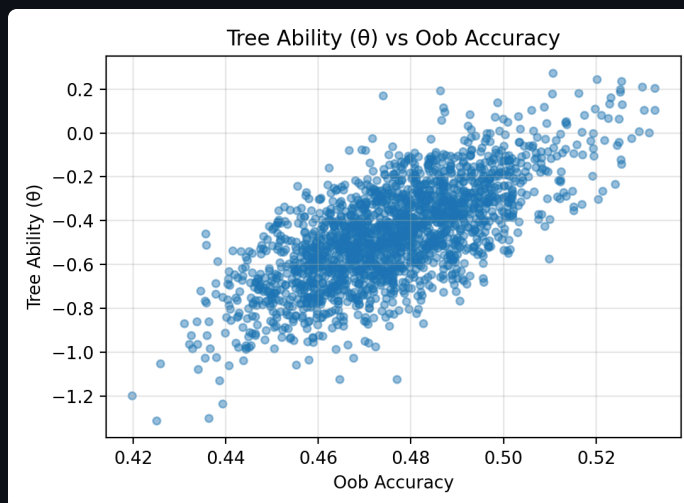
# Tree Attribute Correlations · OOB Accuracy vs $\theta$

- `scripts/analyze_tree_attribute_correlations.py` merges each tree's depth/leaves/OOB stats with  $\theta$  and discrimination aggregates.
- Pearson  $r$  (OOB accuracy,  $\theta$ ): PCA **+0.25**, MobileNet **+0.70**, MNIST **+0.39** — reliable trees earn higher ability across every study.
- CSV/JSON exports: `data/*/tree_attributes_with_signals.csv`, `data/*/tree_attribute_correlations*.json` for deeper dives.

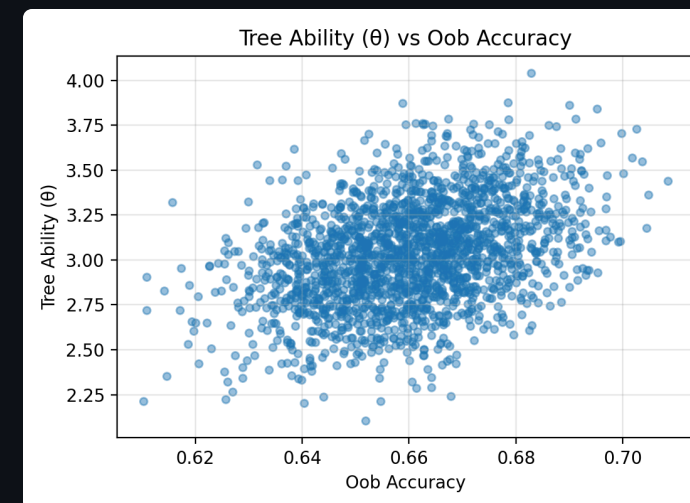
PCA · OOB accuracy vs  $\theta$  ( $r = +0.25$ )



MobileNet · OOB acc vs  $\theta$  ( $r = +0.70$ )



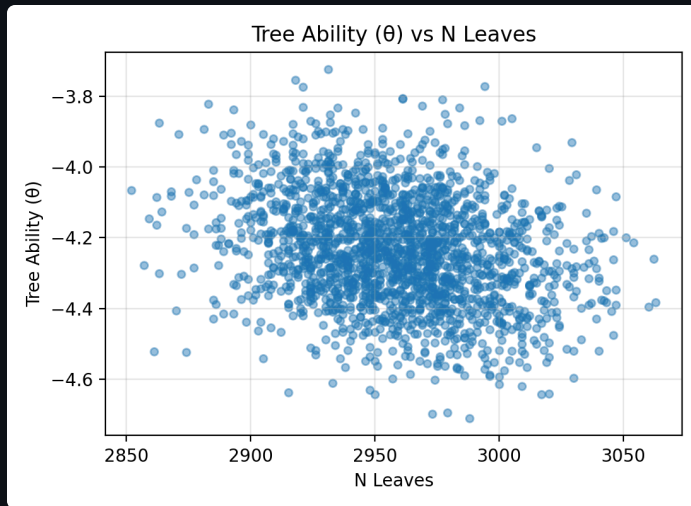
MNIST · OOB accuracy vs  $\theta$  ( $r = +0.39$ )



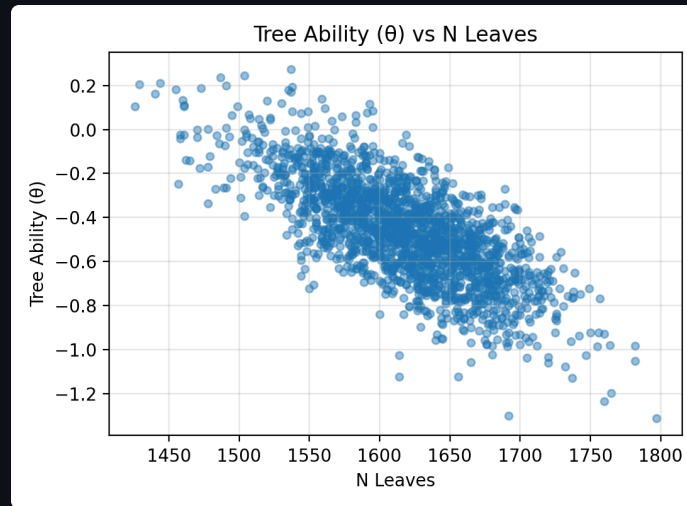
# Tree Attribute Correlations · Leaf Count vs $\theta$

- Pearson  $r$  (leaf count,  $\theta$ ): PCA  $-0.27$ , MobileNet  $-0.73$ , MNIST  $-0.38$  — pruning shallower trees boosts ability rankings.
- Leaf count penalizes overfitting branches; MobileNet shows the steepest drop because high-quality features reward compact trees.

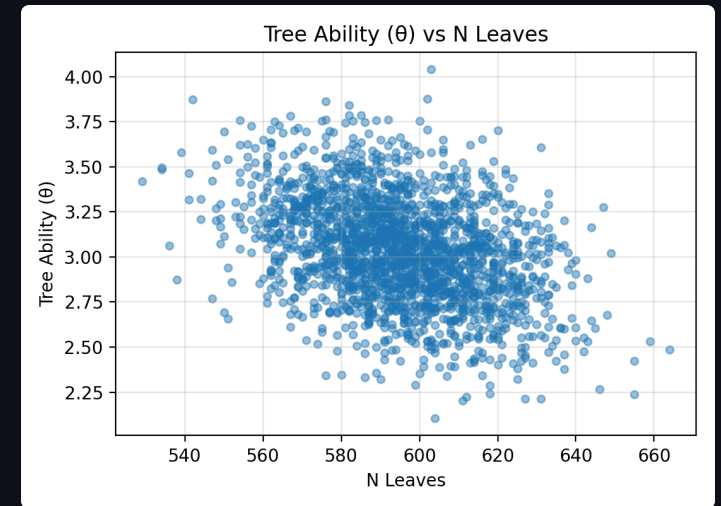
PCA, Leaf count vs  $\theta$  ( $r=-0.27$ )



MobileNet, Leaf count vs  $\theta$  ( $r=-0.73$ )



MNIST, Leaf count vs  $\theta$  ( $r=-0.38$ )



## Key Takeaways

- IRT and RF still move in lockstep:  $\theta$  tracks per-tree accuracy, while  $\delta$  and  $a$  surface stubborn item pockets.
- MobileNet's discrimination tail isolates animal confusions despite stronger features; MNIST flips signs because mistakes are rare.
- 3PL adds a modest guessing floor ( $\sim 0.25$ ) without upsetting  $\theta$ -accuracy alignment.
- Tree attributes expose pruning cues: shallow, high-OOB trees consistently land higher  $\theta$ .

## Next Steps

- Run stability sweeps (50/100 trees, alternate seeds) to quantify variance in  $a$  and  $\theta$ .
- Decide whether 3PL merits extension to PCA/MNIST or documenting as MobileNet-only.
- Finish item-tier overlays (high/medium/low  $a$ ) and align them with the qualitative grids.

## References

- Wilson, M. (2005). *Constructing Measures: An Item Response Modeling Approach*. Lawrence Erlbaum Associates.
- Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and Regression Trees*. Wadsworth.
- Breiman, L. (2001). "Random Forests." *Machine Learning*, 45(1), 5–32.

# Extra Material

# Decision Trees — From Data to Splits

**Idea:** recursively split data to increase *purity* of labels (Breiman et al., 1984).

Example:

“PetalLength < 2.5?” → all *Setosa* left, others right.

| PetalLength | PetalWidth | Species    |
|-------------|------------|------------|
| 1.4         | 0.2        | Setosa     |
| 4.7         | 1.4        | Versicolor |
| 5.5         | 2.0        | Virginica  |

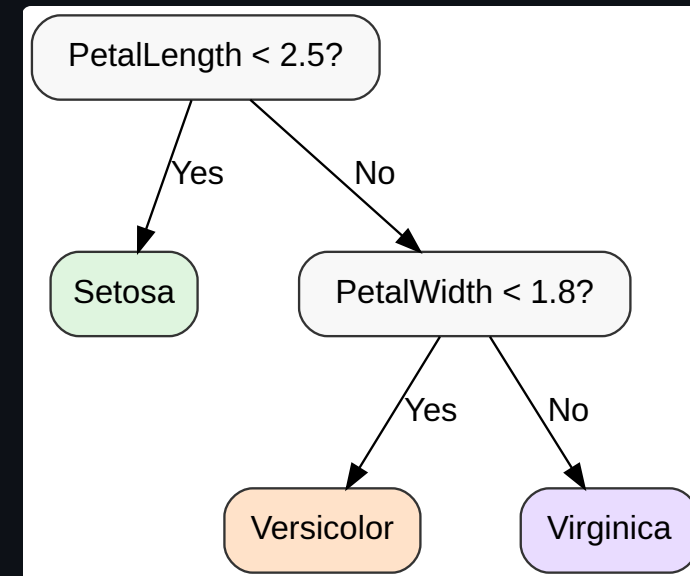
At each node:

- compute **impurity** (e.g., *entropy* or *Gini*):

$$H = - \sum_i p_i \log_2 p_i$$

- choose the split that **maximally reduces impurity** — i.e. makes groups more uniform.

A single tree = a set of *if-then* rules that classify or predict.





# Gini vs. Entropy — Two Lenses on Node Impurity

Entropy (Information Theory):

$$H = - \sum_i p_i \log_2 p_i$$

Measures **uncertainty** — expected information (in bits) needed to classify a random sample. *High when classes are evenly mixed.*

Gini Impurity (Probability of Misclassification):

$$G = 1 - \sum_i p_i^2$$

Measures **chance of error** — probability that two randomly drawn samples from the node belong to different classes.

| Metric  | Theoretical Lens   | Interpretation                | Typical Use                      |
|---------|--------------------|-------------------------------|----------------------------------|
| Entropy | Information theory | “How surprised would I be?”   | ID3, C4.5 trees                  |
| Gini    | Probability theory | “How often would I be wrong?” | CART trees, scikit-learn default |

Both peak when classes are perfectly mixed ( $p = 0.5$ ).  
Gini is slightly flatter — faster to compute, less sensitive to extremes.

# Estimating $\theta$ and $\delta$ — A Toy Rasch Example Worked-Out

We'll fit a **1-Parameter Logistic Model (Rasch Model)** to a tiny response matrix:

| Person | Item 1 | Item 2 | Item 3 | Total $r_i$ |
|--------|--------|--------|--------|-------------|
| A      | 1      | 1      | 0      | 2           |
| B      | 1      | 0      | 0      | 1           |
| C      | 0      | 0      | 1      | 1           |

"1" = correct "0" = incorrect

Three people, three items — small enough to solve by hand.

The Rasch Model:

For each person  $i$  and item  $j$ :

$$P(X_{ij} = 1 | \theta_i, \delta_j) = \frac{1}{1 + \exp[-(\theta_i - \delta_j)]}$$

- $(\theta_i)$ : person's latent **ability**
- $(\delta_j)$ : item's **difficulty**

When  $(\theta_i = \delta_j)$ ,  $(P(X_{ij} = 1) = 0.5)$  — equal odds of success and failure.

# Step 1 — Initial Estimates

A simple starting point uses **logits of proportions**: Compute:

$$\theta_i^{(0)} = \log \frac{r_i/m}{1 - r_i/m}, \quad \delta_j^{(0)} = -\log \frac{s_j/n}{1 - s_j/n}$$

where

- ( $m = 3$ ) items per person
- ( $n = 3$ ) people per item
- ( $r_i$ ) : person totals    ( $s_j$ ): item totals

|        | Formula    | Value  |
|--------|------------|--------|
| Resp_A | logit(2/3) | +0.693 |
| Resp_B | logit(1/3) | -0.693 |
| Resp_C | logit(1/3) | -0.693 |

|        | Formula     | Value  |
|--------|-------------|--------|
| Item 1 | -logit(2/3) | -0.693 |
| Item 2 | -logit(1/3) | +0.693 |
| Item 3 | -logit(1/3) | +0.693 |

Center the item difficulties so  $\sum \delta = 0$ .

## Step 2 — Compute Expected Scores

- Plug into the Rasch model:

$$P_{ij} = \frac{1}{1 + e^{-(\theta_i - \delta_j)}}$$

- Sum across items → expected totals per person.

| Person | Expected $r_i$ | Observed $r_i$ |
|--------|----------------|----------------|
| A      | 1.95           | 2              |
| B      | 1.04           | 1              |
| C      | 1.04           | 1              |

- Pretty close already — the model almost reproduces the data.

## Step 3 — One Newton Update (By Hand)

- Adjust each parameter so model-predicted totals match observed totals:

$$\theta_i \leftarrow \theta_i + \frac{r_i - \sum_j P_{ij}}{\sum_j P_{ij}(1 - P_{ij})}$$

$$\delta_j \leftarrow \delta_j - \frac{s_j - \sum_i P_{ij}}{\sum_i P_{ij}(1 - P_{ij})}$$

- Re-center  $\sum \delta = 0$  after each step.
- After one iteration (rounded):

| Parameter      | Value |
|----------------|-------|
| $\theta_a$     | +0.77 |
| $\theta_\beta$ | -0.75 |
| $\theta_c$     | -0.75 |
| $\delta_1$     | -1.04 |
| $\delta_2$     | +0.52 |
| $\delta_3$     | +0.52 |

## Step 4 — Check Fit

- Recompute expectations:

| Person | Expected $r_i$ | Observed $r_i$ |
|--------|----------------|----------------|
| A      | 1.99           | 2              |
| B      | 1.01           | 1              |
| C      | 1.01           | 1              |

- Perfect alignment — the model now fits.

## What We Learned

- $(r_i)$  (person totals) are **sufficient** for estimating  $\theta$ .
- $(s_j)$  (item totals) are **sufficient** for estimating  $\delta$ .
- Centering fixes the arbitrary origin of the latent scale.
- Precision varies by level: extreme scores  $\rightarrow$  higher uncertainty.

In the Rasch world, **abilities and difficulties calibrate each other** — each  $\theta$  and  $\delta$  defined only in relation to the ensemble.

# From Rasch Estimation to Stochastic Variational Inference (SVI)

Both Rasch estimation and SVI fit **latent-variable models**, but they differ in *how* they estimate hidden parameters.

| Aspect      | Rasch (by hand)                                                                       | Stochastic Variational Inference                                                     |
|-------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Goal        | Find point estimates of $\theta$ and $\delta$ that best reproduce observed responses. | Approximate full posterior distributions over latent variables.                      |
| Computation | Deterministic updates (e.g., Newton–Raphson).                                         | Stochastic gradient ascent on an <i>evidence lower bound</i> (ELBO).                 |
| Uncertainty | Single best estimate per parameter.                                                   | Explicitly models uncertainty via variational distributions ( $q(\theta, \delta)$ ). |
| Scale       | Works for small datasets, exact likelihood.                                           | Scales to millions of observations using minibatches.                                |
| Analogy     | Matching observed vs. expected totals until equilibrium.                              | Minimizing the KL divergence between approximate and true posteriors.                |

Rasch estimation is like hand-tuning a few dials until predicted counts match reality.

SVI automates that process with noisy gradients — learning not just *where the dials land*, but *how uncertain we are about their true positions*.